

Effective line redistribution in kilonova ejecta: transport-relevant structure is frequency-local

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A reduced model of a high-dimensional microscopic process is usually judged by how well it reproduces the microscopic dynamics it replaces. We show, for lanthanide line fluorescence in kilonova ejecta, that this criterion selects the wrong model. Holding the atomic physics, the resolved line opacity and the physical state fixed, we compare two reduced representations of the redistribution of absorbed energy across frequency: a coarse graining into contiguous frequency groups, and a global non-negative factorisation with the same number of archetypal exit distributions and many more free parameters. At 32 archetypes on Nd II the global representation reproduces the microscopic redistribution events 1.8 times better and the emergent broad-band light 20.8 times worse. Fidelity to the microscopic distribution is not the criterion that selects the representation the radiation samples. The local operator reaches a 0.10 mag photometric criterion with a handful of groups where the scalar thermalisation closure, tuned to the same criterion, misses by 0.82 and 1.43 mag; it remains compact across gas temperature, density, illumination and a realistic 13-ion mixture, where 4 groups suffice and the scalar closure misses by 1.55 mag; but it is state dependent, interpolable along single state coordinates while the one coupled trajectory tested required a refit, and for one ion its required resolution changes with the state.

INTRODUCTION

When a process with millions of microscopic degrees of freedom is replaced by a reduced model, the reduction is normally built and judged against the microscopic dynamics: the reduced model that reconstructs the fine-grained process most accurately is taken to be the best. Model reduction aimed at a quantity of interest recognises that the appropriate reduced model can depend on what one wants from it [1, 2], and reduced and low-rank formulations of radiative transfer exist [3]. The question we address is narrower and physical: in a concrete transport problem where the microscopic process can be simulated without the reduction under study, which of its degrees of freedom actually reach the observable radiation, and does fidelity to the microscopic process identify them?

The process is lanthanide line fluorescence in the ejecta of a neutron-star merger. The open f -shells of the lanthanides carry of order 10^6 transitions per ion at the relevant temperatures [4, 5]; a photon absorbed in one line is re-emitted, after a cascade through the ion’s levels, in another, and this redistribution of energy across frequency is a major mechanism of non-local redistribution in wavelength in a kilonova’s light. High-fidelity transport keeps the identity of every transition for exactly this reason: line-by-line calculations with tens of millions of lines follow the absorption and the cascade explicitly, through a stochastic macroatom [6–8]. Cheaper calculations replace the cascade by a parameterised redistribution — a scalar thermalisation probability ϵ that sends a fraction of the absorbed energy to a thermal pool and scatters the rest coherently, or a frequency-grouped treatment of the opacity with a prescribed re-emission within each group

[9]. Frequency redistribution as a formalism is classical [10]; what is not known is how much of the fluorescence response a transport calculation needs, and what organises the part it needs.

We hold the opacity at the resolved Sobolev value of every line, take an energy-conserving downward macroatom as the unreduced reference, and build reduced operators from the reference’s own events on one set of random seeds, scoring them on another. Three preregistered gates then ask, in order, whether a compact operator exists where the scalar closure fails; whether a representation that keeps contiguous frequencies together reaches the observables with no more archetypal exit distributions than a global low-rank one; and whether that structure survives changes of state and composition — the three results sections below. The second gate is the result of this Letter. At matched archetype count and at the high end of that count, the representation that reproduces the microscopic redistribution better reproduces the light worse. In lanthanide fluorescence transport, generic fidelity to the microscopic redistribution does not identify the degrees of freedom that control the emergent radiation; a frequency-local representation does.

SET-UP

Every calculation is a single-zone transport through the photospheric shell of a published lanthanide-rich ejecta model at two days, with the shell’s own density, gas temperature and expansion time and a black-body core illuminating it. One ion at a time — La II, Ce II or Nd II, at its number density in the shell, with levels and lines from a calibrated atomic data set [11] and populations in local

thermodynamic equilibrium — carries the opacity; La II is thin enough at its partial density that every closure agrees, and is a control. Packets carry indivisible energy. The reference is a downward macroatom: on absorption the packet walks the radiative cascade with branching probabilities $A\beta$ until it exits in a line whose photon escapes, conserving energy at every step; the identity between injected and accounted energy is verified to 10^{-10} on every run that enters a reading (End Matter).

The reduced operator is built from the reference's events (absorbed frequency, exit frequency, energy before and after). On N_g log-spaced frequency groups, R_{ij} is the fraction of energy absorbed in group i that exits in group j ; within each output group the exit line is drawn from a discrete table of the distinct exit lines the events populated, weighted by energy. Transport with the operator replaces the cascade by one draw from R_{ij} and one from the table. Operators are built on three seeds and scored on three disjoint seeds. The global family of the locality test replaces the transition matrix by a rank- k non-negative factorisation of the fine-grouped matrix on the same exit tables; the state-dependent operators of the generality test are rebuilt at, transported to, or interpolated between states as described there. Every closure is scored by its maximum absolute error in the broad-band magnitudes and colours over the bands that carry at least one per cent of the luminosity, by an integrated spectral distance, and by an energy-weighted total-variation distance to an independent fine-grouped matrix from the evaluation events; the criterion throughout is 0.10 mag in band and 0.10 mag in colour. The gates, their readings and the amendments made during the runs are in the End Matter.

A COMPACT OPERATOR EXISTS WHERE A SCALAR CLOSURE FAILS

The scalar closure at its minimax ϵ — the grid value that minimises the same joint band-and-colour statistic the operator is judged by — misses the reference by 1.43 mag on Ce II and 0.82 mag on Nd II (the preregistered ϵ^* , which minimises the mean band error, misses by 1.40 and 1.06 mag; End Matter); the group operator meets the criterion at $N_g = 16$ and $N_g = 2$ (Fig. 1a). The compression is not an artefact of one cascade prescription: replacing the reference by a radiation-field-driven macroatom with internal upward transitions moves the photometry by 2.09 mag on Ce II, and an operator rebuilt from that reference follows it to 0.026 mag while the operator trained on the original reference stays 1.96 mag away (End Matter, Fig. 3). The operator carries the fluorescence physics it is built from. This establishes that there is a non-trivial reduced model to explain; the explanation is the next section.

OBSERVABLE FIDELITY IS NOT MICROSCOPIC FIDELITY

Two families of operator were built on identical exit tables and identical opacity, differing only in the group-to-group matrix: local coarsening into N_g contiguous frequency blocks, and a global non-negative factorisation into k archetypal exit distributions mixed freely for every fine input group. The matched axis is the number of archetypes. At equal count the global family is far the more expressive, with 8,192 free numbers against 1,024 at 32 archetypes. The factorisation minimises a Frobenius reconstruction error of the fine matrix, not the energy-weighted total-variation loss reported here, so its microscopic advantage is measured, not built in: it holds at the high end of the ladder and not at every count. If the compression of the first results section reflected a globally low-rank redistribution law, the global family would reach the photometric criterion with fewer archetypes.

It does not. On Ce II the local operator passes at 16 archetypes and the global one at 32; on Nd II the local operator passes at 2 and no member of the global family passes at any rank tried (its count is undefined). The contrast is sharpest at the high end of the ladder, where both families are well resolved. At 32 archetypes on Nd II the global operator's event-level total-variation loss is 0.075 against the local operator's 0.132, and its photometric error is 0.162 mag against 0.008 mag:

$$E_{\text{micro}}^{\text{global}} < E_{\text{micro}}^{\text{local}}, \quad E_{\text{obs}}^{\text{global}} > E_{\text{obs}}^{\text{local}}, \quad (1)$$

by factors of 1.8 and 20.8 respectively, and the same inversion holds on every ion at that count (Fig. 1b–d; colours in Table II). It is a statement about the high-count end of the ladder, not about every rank: at the local operator's own passing count the global family fits the events better on Ce II (16 archetypes) and does not on Nd II (2), where its event loss is the larger. The degrees of freedom that minimise a generic distance between microscopic redistribution distributions are not the degrees of freedom that carry the radiation field. The operator that keeps contiguous frequencies together reproduces the light with a few blocks; the factorisation, free to mix distant frequencies, does not, however well it reconstructs the events at the counts where it does. Whether adjacency itself carries the transport-relevant structure, rather than the rank bound or the parameter count the two families do not share, is the subject of a preregistered ablation of the frequency ordering at fixed block count (End Matter).

The compression is two-level. The reduced model as implemented is $P_{\text{red}}(\nu_{\text{out}} | \nu_{\text{in}}) \approx P(j | i) P(\nu_{\text{out}} | j)$ with $i = i(\nu_{\text{in}})$: the group-to-group factor is extremely coarse, and once the output group is chosen the exit line is drawn from that group's table, which no longer remembers which input group produced it. The second factor still needs a sparse but real set of exit lines: truncating

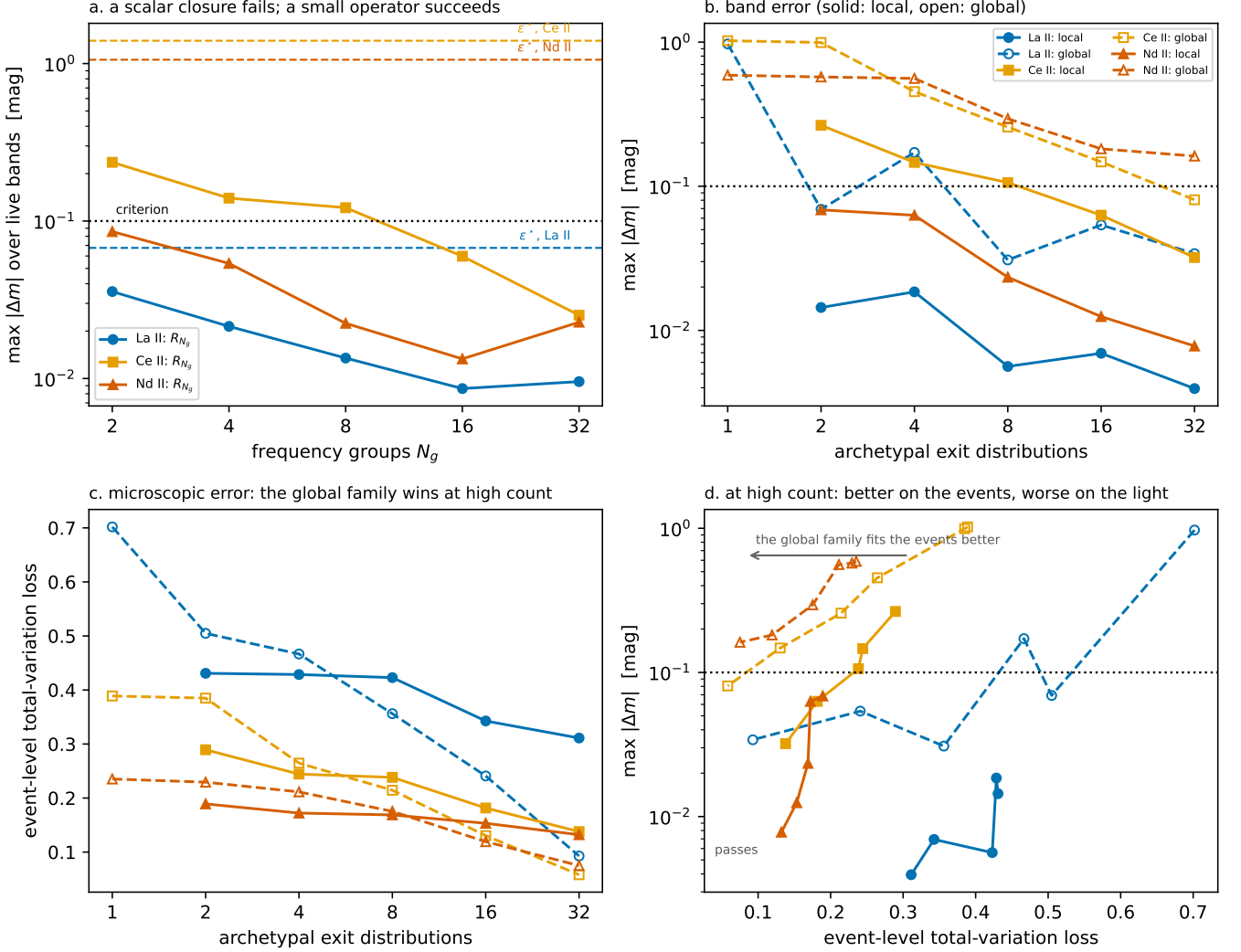


FIG. 1. **Better on the microscopic events, worse on the light.** (a) A scalar closure fails where a small operator succeeds: maximum photometric error against the energy-conserving downward macroatom for the group operator at $N_g = 2 \dots 32$ (solid) and the scalar closure at its preregistered ϵ^* (dashed); dotted, the criterion. (b) Band error and (c) event-level total-variation loss against the number of archetypal exit distributions, for local frequency coarsening (solid) and the global factorisation (open). (d) The two against each other: at the high-count end of the ladder the global points move left, fitting the microscopic redistribution better, while staying high, reproducing the emergent light worse. Panels (b)–(d) show the band error; the criterion also bounds the colour error, and the contrast quoted in the text is the 32-archetype point.

the tables by retained energy, the light needs 0.358 of the distinct exit lines on Ce II and 0.113 on Nd II at the same thresholds. We do not claim that the atomic network has only a handful of degrees of freedom. We claim that microscopic reconstruction error does not find the part of it the transport needs, and that a representation keeping contiguous frequencies together does.

A STATE-DEPENDENT FAMILY, COMPACT THROUGHOUT

Is the local operator a property of the ion or of the state it was built in? We moved one coordinate at a time from

the original state along 4 axes — gas temperature, number density, the temperature of the illuminating source, and the physical trajectory at one, three and five days along which all of them move together — and at each of 14 states per ion scored, against that state's own reference, both the original operator transported unchanged and an operator rebuilt there (Fig. 2), both at 16 groups. The transported operator is not predictive: it fails on every axis for Ce II and on all but the source-spectrum axis for the other two ions. The rebuilt operator passes at every readable state of Nd II and La II, with at most 4 groups needed anywhere on Nd II. There is no universal R_{ij} ; the failure of transfer is not a failure of compression.

What varies is the content of a small operator, and it varies smoothly along single coordinates. At the interior point of each axis an operator interpolated linearly in the logarithm of the moved coordinate between the two bracketing states — transition rows, deposition channel and exit tables on the union of the endpoints’ exit lines, never using the interior state’s events — passes the criterion on the temperature, density and source axes for every ion, reducing Nd II’s error at the interior temperature from 0.150 to 0.032 mag. The useful object is a tabulated family $R(\theta)$. Its boundary is the coupled trajectory: at three days on Nd II the transported operator misses by 0.181 mag, the interpolant by 0.117 mag, and only the operator rebuilt there passes, at 2 groups with 0.013 mag. The one-dimensional interpolation that succeeds along independently varied coordinates does not predict the tested coupled trajectory; the present calculation does not distinguish multivariate coupling from insufficient sampling or an inadequate interpolation coordinate. Ce II adds a second qualification: its rebuilt 16-group operator fails at 7 of 14 states, on colour, by at most 0.062 mag, and a 32-group operator passes at every one. Compactness survives; a fixed representation size does not.

Composition closes the argument. Three single-ion operators combined by an opacity-share rule, with no fit to the blend, do not reach the criterion at any valid resolution (0.142 mag at best), while an operator built on the three-ion blend passes at 4 groups: per-species operators do not compose. Yet on the full 13-ion composition of the benchmark state — the mixture a transport calculation actually needs — the operator built on the blend passes at 4 groups (0.088 mag in band, 0.073 in colour) while the scalar closure at its minimax ϵ misses by 1.55 mag, the widest scalar failure in this work. The combined physical system is itself highly compressible even though it cannot be assembled from its parts. This demonstrates compression of that mixture at the tested state; predictive transfer of a mixture-specific operator to other states has not been demonstrated.

CONCLUSIONS

Of the order 10^6 transitions of a lanthanide ion, the transport sees a fine distribution of redistribution events; of that distribution, the emergent light sees a small frequency-local operator; and most of the information between the two is discarded without harm. The decisive evidence is the inversion of Eq. (1) at 32 archetypes: a representation that reproduces the microscopic redistribution better reproduces the light worse, so the microscopic distribution is the wrong target for a reduced model of this process. The representation the light accepts keeps contiguous frequencies together; whether adjacency itself, rather than rank or parameter count, is

what it exploits is the question the preregistered ordering ablation (End Matter) settles. The local operator is not universal — it is a state-dependent family, interpolable along single coordinates, refit on the one coupled trajectory tested, with a resolution that is itself state-dependent for Ce II — and it does not compose from single species; but on the realistic mixture, at the one state tested, it is as compact as on any single ion. It is built from the reference’s own events, so it does not yet remove the reference calculation for a new state or mixture; its practical form is a precomputed table $R(\theta)$, plausible along single coordinates and not yet established generally. The limitations are those of the set-up: single-zone, time-independent, with the radiation field imposed and its influence measured only through the source-spectrum axis, on one coupled path. None of them touches the inversion, which is measured on identical events, identical opacity and identical observables. Transport can discard most of the microscopic fluorescence information — but only if the reduced coordinates respect the structure the radiation actually samples.

END MATTER

The gates. Each of the three questions was preregistered as a gate — the state, the grids, the metrics, the thresholds, the gray conditions that invalidate a reading and the readings themselves fixed and committed before the run — and read gray-first. Table I gives the per-ion readings of the first gate (the group count at which the operator meets the criterion, the preregistered scalar ϵ^* with its maximum band error, and the minimax ϵ_{mm}^* of the main text with the joint statistic) and of the second (the archetype counts of the local and global families, and the fraction of exit lines the light needs), with the robustness check against the radiation-field-driven macroatom; Table II the microscopic and observable errors of Eq. (1) at 32 archetypes on every ion; Table III the third gate per ion — its three readings, the axes along which the transported operator stays predictive, how often the rebuilt operator fails and the largest resolution it needed, and the interpolation error at each single-coordinate interior point, whole operator against transition matrix alone. The third gate’s first reading is Red for Ce II (the resolution result above) and its second Red for Nd II (the trajectory); its mixture readings are Yellow for composability and Green for the realistic blend.

The scalar comparator. The preregistered ϵ^* is the grid value minimising the mean band error, reported by its maximum band error; the operator is judged by the maximum band and colour error at one threshold. The main text therefore quotes the scalar at ϵ_{mm}^* , the grid value minimising that same joint statistic, recomputed from the existing grids: 0.10 on Nd II (0.82 mag against 1.06 mag at ϵ^*), 0.00 on Ce II (1.43 mag), and 0.05 on the

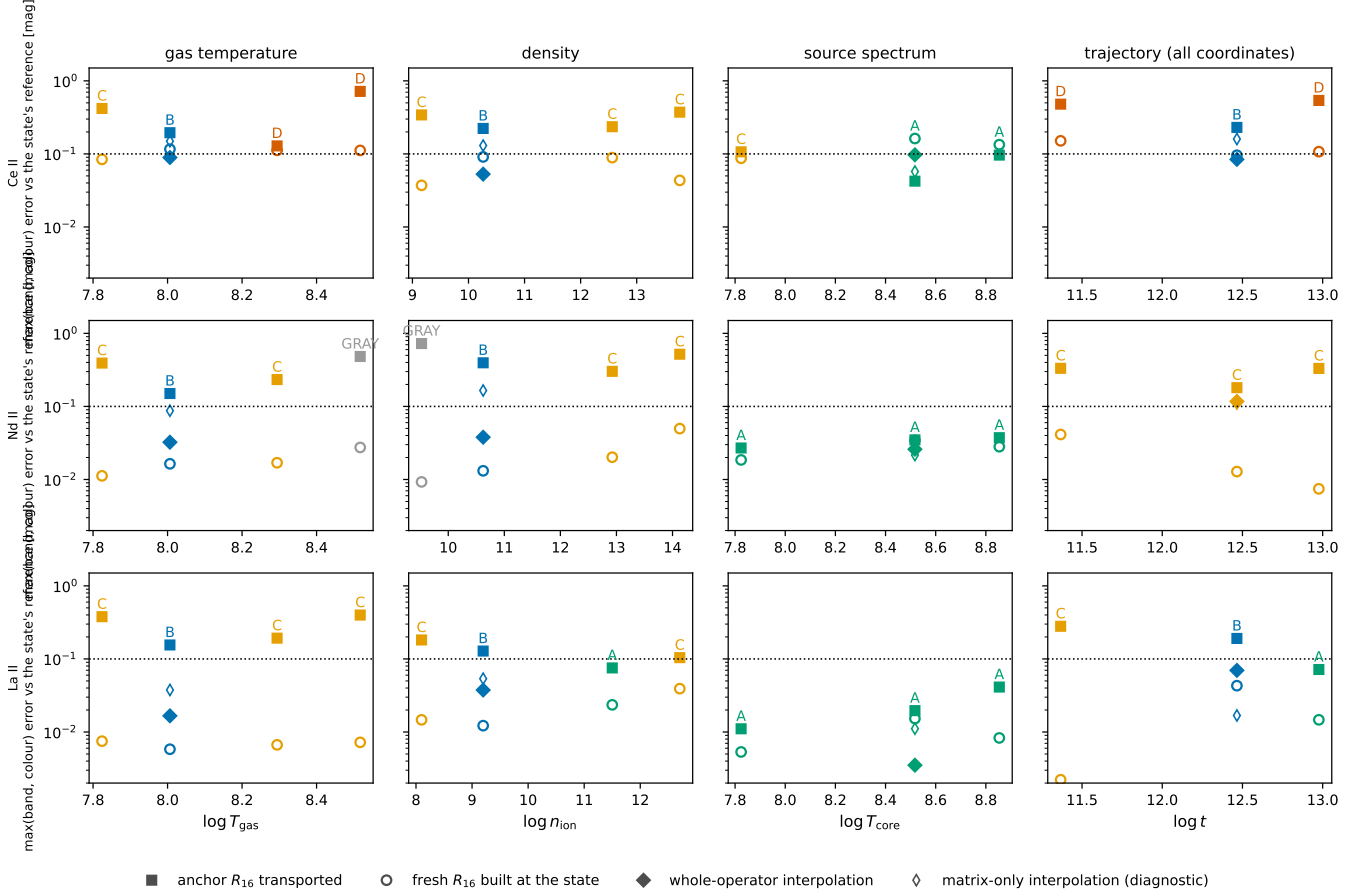


FIG. 2. **Transfer is not compression.** The decision statistic — the larger of the maximum band error and the maximum colour error — against the state’s own reference for the original operator transported unchanged (filled squares), the operator rebuilt at the state (open circles), and at each axis’ interior point the whole-operator interpolant from the two bracketing states (filled diamonds; matrix-only interpolant, open diamonds). Rows: Ce II, Nd II, the La II control; columns: gas temperature, density, source spectrum, and the trajectory along which all move together. Dotted: the criterion, on the same statistic; the letter is the preregistered reading of each state (A transported operator passes; B only the interpolant; C only a fresh fit; D none at 16 groups). Grey: states excluded by the preregistered validity checks.

13-ion blend (1.55 mag against 1.69 mag). No reading changes.

Cost. Table IV separates what the operator costs. Offline, the reference macroatom is run on the build seeds and the operator is built from its events. Stored, the operator is an $N_g \times N_g$ matrix of a few hundred bytes and discrete exit tables of up to a few megabytes — the tables, not the matrix, are the object. Online, transport with the operator costs 0.94–1.15 of transport with the downward macroatom at the same seeds and packets, because one draw from R_{ij} and one from a table replaces a cascade that is itself cheap once the line list is in memory; the scalar closure at $\epsilon = 0.1$ costs 4.1 to 8.8 times the reference because coherent scattering multiplies the interactions per packet. The operator therefore does not yet remove the reference calculation for a new state or mixture; its practical path is a precomputed table $R(\theta)$, which the third gate shows to be plausible along single

coordinates and not yet established generally.

Preregistered controls in progress. Two were preregistered on the referee’s reading and are not run at the time of this draft. The ordering ablation takes the frozen fine operator of the second gate, permutes its frequency indices, applies the same block coarse-graining and undoes the permutation, so that the block count, rank bound, parameter count, exit tables, events, seeds and transport are unchanged and only physical adjacency is destroyed; thirty-one fixed random orderings against the physical one, on Ce II and Nd II at every block count of the ladder, read by the rank of the physical ordering. The paired-seed check re-transportes every decision leg of the third gate whose error lies within a preregistered window of the threshold on twelve common seeds, with the trained operators preserved, and reports a paired bootstrap interval on the joint statistic that decides each letter.

Amendments during the runs. Three were made and are recorded in the archived preregistration with what had been seen when each was made. In the second gate, two implementation defects in computing the factorisation (a live row sent to zero; a fixed iteration count that had not converged) were found on the control ion and fixed before any decisive ion's numbers were read, and a validity control that compared the local family on coarse and fine exit tables fired on both decisive ions and was answered by its preregistered remedy, transporting the local family on the shared fine tables; no threshold changed. In the third gate, an unclamped cumulative-sum sampler overflowed on the 13-ion mixture, the largest line list in this work; it was fixed with the clip its sibling samplers already carried, its reachability in every earlier record was established by inspection (any overflow crashes, so no completed record could have been affected), and the mixture calculation was repeated. A separate, non-reproducible corruption of one energy-accounting sum in three runs was audited by rerunning each, found not to affect the photometry, and answered by excluding any run with an invalid energy ledger from every reading.

Transport and validation. The transport is a Monte Carlo Sobolev code with worldline relativity and energy packets. Its extended validation on coarse line-opacity treatments is the subject of a companion methods paper (in preparation); the particular tests this Letter depends on are documented in the public record cited below and are independent of that paper. Every run that enters a reading closes the identity between injected and accounted energy to 10^{-10} ; a run whose ledger fails that check is excluded from every reading, as the amendments paragraph records, and the record carries each run's ledger. The resolved Sobolev transport is pinned against a deterministic formal-solution reference on identical rays, the downward macroatom reproduces an explicit level-by-level cascade's energy per line digit for

digit, and every operator is built on seeds disjoint from those it is scored on, the build and evaluation records both archived. The generality test lays every kernel of an ion on one frequency support frozen over the whole state domain, so that an operator transported between states is never clipped to a different range, and the clipped fraction is recorded per state.

Data availability. The transport code, the preregistrations of the three gates, every run record, the frozen record from which every number in this Letter is generated, and the scripts that regenerate the numbers, tables and figures are archived at github.com/zhuygln/revisit_sobolev (tags `paperB-g2-prereg`, `paperB-g2-freeze`, `paperB-g3-prereg`, `paperB-g3-freeze`).

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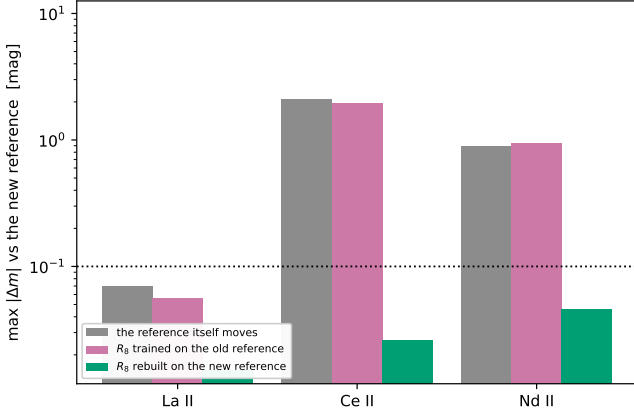


FIG. 3. **The operator follows the fluorescence physics it is built from.** When the reference is replaced by a radiation-field-driven macroatom with internal upward transitions, the reference itself moves (grey); an operator rebuilt from the new reference's events follows it (green), while the operator trained on the previous reference does not (pink). Dotted: the criterion.

TABLE I. The first two gates, per ion.

ion	N_g^*	ϵ^*	error at ϵ^*	ϵ_{mm}^*	$E_{\text{joint}}(\epsilon_{\text{mm}}^*)$	K_{local}^*	K_{global}^*	L^*	R_8 vs the new reference
			[mag]		[mag]				[mag]
La II	2	0.20	0.07	0.20	0.07	2	2	0.504	0.015
Ce II	16	0.00	1.40	0.00	1.43	16	32	0.358	0.026
Nd II	2	0.00	1.06	0.10	0.82	2	undefined	0.113	0.046

TABLE II. Microscopic against observable error at 32 archetypes, in band and in colour.

ion	m_{event}	global	m_{event}	local	max $ \Delta m $	global	max $ \Delta m $	local	max $ \Delta c $	global	max $ \Delta c $	local
					[mag]		[mag]		[mag]		[mag]	
La II		0.093		0.311		0.034		0.004		0.017		0.006
Ce II		0.058		0.138		0.081		0.032		0.034		0.051
Nd II		0.075		0.132		0.162		0.008		0.180		0.010

TABLE III. The third gate, per ion: the readings, the axes along which the transported operator stays predictive, how often the rebuilt 16-group operator fails and the largest resolution it needed, and the interior-point interpolation error (whole operator / transition matrix only) on the three single-coordinate axes, in mag.

ion	C1	C2	C3	transfer holds	fresh R_{16}	fails	max K_{rec}^*	interp. T	interp. D	interp. J
				(axes)	(of states)			whole / matrix	whole / matrix	whole / matrix
La II	GREEN	YELLOW	GREEN	J	0/ 14		2	0.010/ 0.020	0.030/ 0.032	0.003/ 0.011
Ce II	RED	YELLOW	YELLOW	none	7/ 14		32	0.064/ 0.099	0.047/ 0.130	0.050/ 0.052
Nd II	GREEN	RED	GREEN	J	0/ 12		4	0.032/ 0.087	0.034/ 0.110	0.026/ 0.022

TABLE IV. What the operator costs, from the records: the build reference run (three build seeds at the stated packets) and its events, the stored operator at the gate's group count (matrix against exit tables), and the wall time of the reference, the operator and the scalar closure at $\epsilon = 0.1$ on the same evaluation seeds and packets.

record	build run	events	N_g	matrix	exit lines	operator	reference / operator	scalar $\epsilon = 0.1$
	[s]			[B]		[kB]	[s]	/ reference
La II	1	322,994	2	32	524	16	1/ 2	4.2×
Ce II	12	5,439,289	16	2,048	12,510	303	11/ 13	4.4×
Nd II	128	18,641,032	2	32	86,187	2023	127/ 134	4.1×
13-ion blend	193	30,634,068	4	128	175,086	4107	187/ 176	8.8×